Did the Ads Actually Work?

Pages

- Overview
- Targeting & money
- How to measure

Red marks the one thing to look at; blue carries the data.

Independent analysis of public data (Criteo Uplift v2.1, CC BY-NC-SA 4.0; Iowa Liquor Sales). Not affiliated with or endorsed by Criteo, Google or the State of Iowa.

Did the ads actually work?

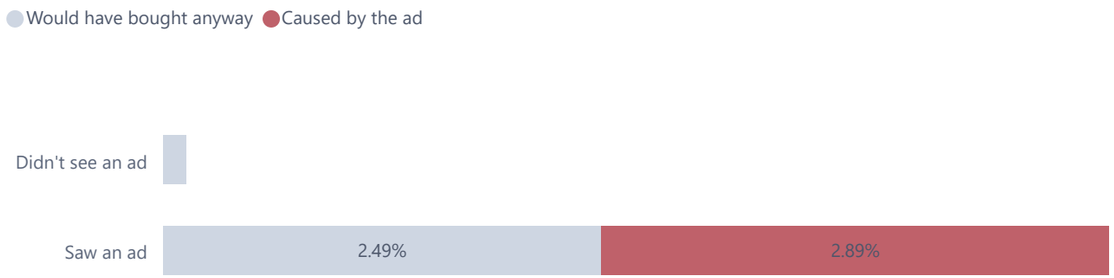
An estimated 54 of every 100 conversions credited to retargeting (view-through) were caused by it.

Randomised test · 13,979,592 users · adjusted for a small leak

Caused per 100 credited (estimated)	Seeing an ad: dashboard view	Seeing an ad: estimated causal effect	Conversion lift, whole treated group
54	41x	2.2x	+52%
95% interval 50 to 58	Saw an ad vs didn't, as a dashboard shows	From the randomised test (CACE)	Adjusted (raw +59%)

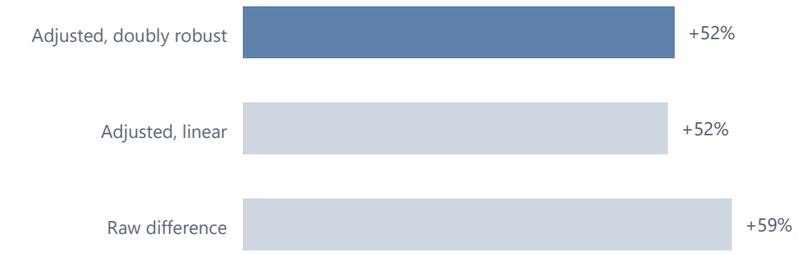
Seeing an ad looks like a 41x boost. The estimated causal effect is 2.2x

Conversion rate · users who saw an ad were already 19x likelier to buy; the test separates that from the ad



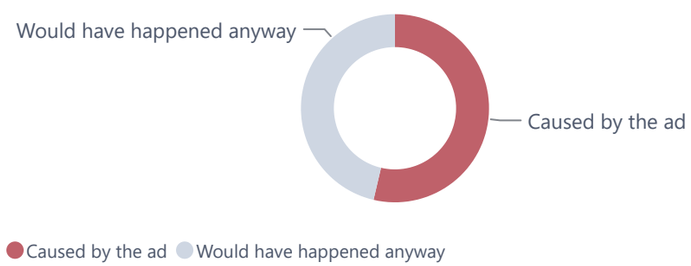
Correcting the leak: conversion lift +59% raw, +52% adjusted

Whole treatment group vs control · preferred: adjusted for pre-treatment features; 7 adjusted specifications give +50% to +56%



Ads caused an estimated 54 of 100 credited conversions

View-through credit · 95% interval 50 to 58; 51 to 58 across adjustment methods · attribution overstates return about 1.9x



How to read this page

Criteo randomly blocked ads for 15% of users. Any difference between the groups was caused by the ads.
Users who saw an ad converted at 5.38%; without it they'd have converted at about 2.49%. So ads work, but far less than a dashboard suggests.
The test had a small leftover imbalance from merging several advertisers' tests. Headline numbers are adjusted for it, assuming the 12 pre-treatment features capture it; raw figures are shown for comparison.
Estimates describe this benchmark file (Criteo sub-sampled it), not every retargeting campaign.
→ Judge retargeting on what it causes: incremental ROAS = dashboard ROAS × caused share.

FINDING	EVIDENCE	INTERPRETATION	NEXT ACTION
An estimated 54 of every 100 credited conversions were caused by the ads.	Seeing an ad lifts conversion from 2.49% to 5.38%: 2.2x, not 41x.	Attribution credits ads with keen shoppers' sales; return is overstated about 1.9x.	Report incremental ROAS next to attributed ROAS in every campaign review.

Source: Criteo Uplift v2.1 (CC BY-NC-SA 4.0) · randomised test, 13,979,592 users · levels describe Criteo's sub-sampled benchmark file

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Who to target, and does it pay?

Half the spend, aimed at likely buyers, keeps 84% of caused conversions; in the scenario, the weaker half loses money.

2.8M held-out users · money is scenario analysis (ROAS 8, margin 30%)

Effect in the top 20% of users
91%
Of the ads' effect on conversions

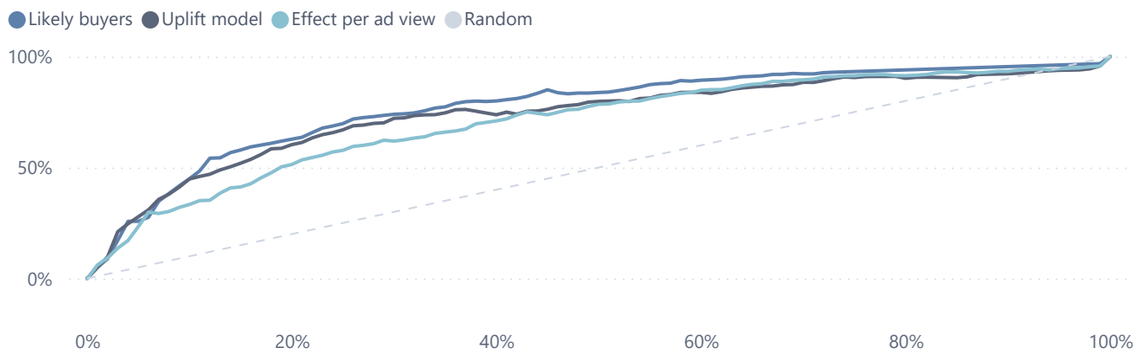
Half the spend keeps
84%
Aimed at likely buyers; 95% interval 78% to 90%

Incremental profit per £1 (scenario)
+£0.29
Dashboard implies +£1.40 (scenario)

Weaker half: profit per £1 (scenario)
-£0.58
The half of spend on the least likely buyers

Half the ad spend, aimed at likely buyers, keeps 84% of the conversions ads cause

Share of the ads' effect on conversions captured vs share of spend (ad views) used, best users first



Scenario: incremental profit per £1 of ads, all spend

By assumed dashboard ROAS (rows) and margin (columns) · red = loss

Dashboard ROAS	20%	30%	40%	50%
4	-£0.57	-£0.36	-£0.14	+£0.07
6	-£0.36	-£0.03	+£0.29	+£0.61
8	-£0.14	+£0.29	+£0.72	+£1.15
10	+£0.07	+£0.61	+£1.15	+£1.68
12	+£0.29	+£0.93	+£1.58	+£2.22
15	+£0.61	+£1.41	+£2.22	+£3.02

The weaker half of spend loses money in 21 of 24 combinations

Scenario: incremental profit per £1 on the half of spend aimed at the least likely buyers · red = loss

Dashboard ROAS	20%	30%	40%	50%
4	-£0.86	-£0.79	-£0.72	-£0.65
6	-£0.79	-£0.69	-£0.58	-£0.48
8	-£0.72	-£0.58	-£0.44	-£0.30
10	-£0.65	-£0.48	-£0.30	-£0.13
12	-£0.58	-£0.37	-£0.17	+£0.04
15	-£0.48	-£0.22	+£0.04	+£0.30

Break-even dashboard ROAS: 6.2 at a 30% margin

The weaker half needs 19.2 · no ROAS assumption needed

Margin	all spend	best half	other half
20%	9.3	5.6	28.7
30%	6.2	3.7	19.2
40%	4.7	2.8	14.4
50%	3.7	2.2	11.5

FINDING
A fifth of users hold 91% of the effect; half the spend keeps 84%.

EVIDENCE
Uplift models graded on 2.8M held-out users by spend; a simple likely-buyer model matched them.

INTERPRETATION
In the scenario the weaker half makes -£0.58 per £1; a hypothesis, not a proven saving.

NEXT ACTION
Test it: a 50/50 holdout in that segment needs about 0.4 million users.

Source: Criteo Uplift v2.1 (CC BY-NC-SA 4.0) · randomised test, 13,979,592 users · levels describe Criteo's sub-sampled benchmark file · money uses stated assumptions · spend measured as ad views

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How to measure it next time

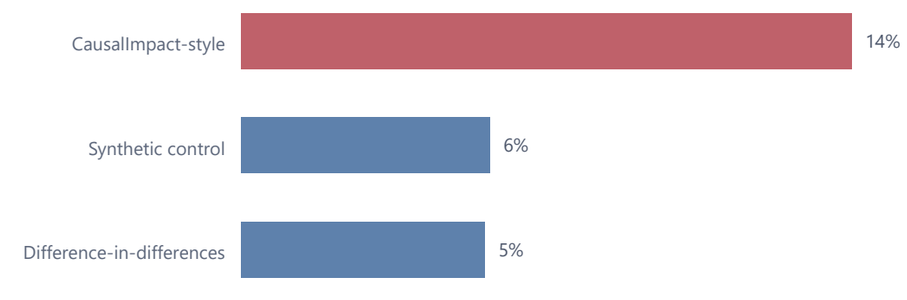
User-level holdouts see small effects; geo tests on noisy sales only catch big ones.

Geo: methods benchmark, 2,700 simulated campaigns on real Iowa sales

Users for a +10% conversion lift	Smallest lift a geo test catches	Difference-in-differences false alarms	CausalImpact-style false alarms
4.9M	9%	5%	14%
1.8 million with a 50/50 split	Best design: 20 counties for 12 weeks	3% to 8% across designs; promised 5%	10% to 18%: cries wolf

The CausalImpact-style model cries wolf 2 to 3.5 times too often

Share of no-campaign tests flagged as significant, average of 9 designs · promised: 5%



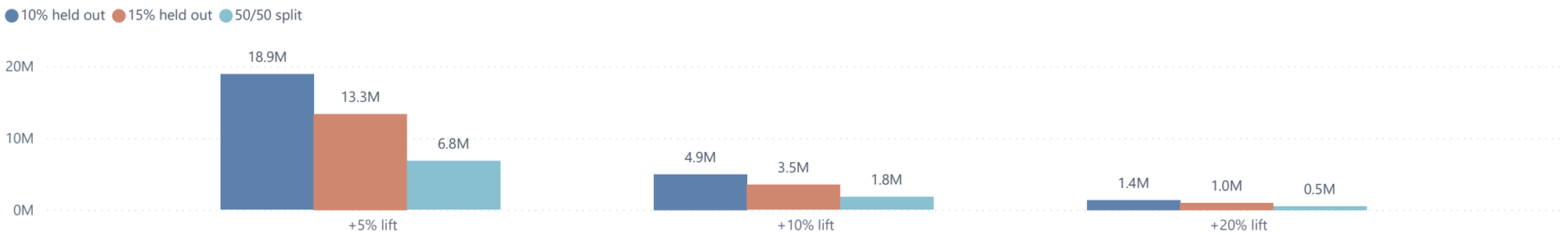
Even 20 counties for 12 weeks only reliably detect about 9%

Difference-in-differences: smallest lift detected 80% of the time · real Iowa sales, simulated campaigns

Test design	4 weeks	8 weeks	12 weeks
5 counties	23%	19%	16%
10 counties	20%	13%	13%
20 counties	12%	10%	9%

Rare conversions need big holdouts: about 4.9 million users to see a +10% lift with a 10% holdout

Users needed to detect a conversion lift 80% of the time (5% false-alarm rate) · 1.8 million with a 50/50 split · base rate from the Criteo control group



FINDING
Holdouts see small effects; geo tests on noisy weekly sales only catch big ones.

EVIDENCE
Best geo design detects about 9%; CausalImpact-style raised 10% to 18% false alarms.

INTERPRETATION
Rare conversions need scale: 4.9 million users for a +10% lift.

NEXT ACTION
Keep an always-on holdout; geo tests only where holdouts are impossible, sized by simulation.

Sources: Iowa Liquor Sales (BigQuery public data), real sales with simulated campaigns · holdout sizing from Criteo control-group rates